End-to-End Project Report: Customer Behavior Analysis

Executive Summary

This project analyzes the shopping habits of **3.9K customers** to uncover actionable business insights. The end-to-end analytical workflow involved data cleaning and preparation using Python, exploratory data analysis using SQL Server, and data visualization using Power BI. The analysis provides a comprehensive look into customer demographics, category performance, and purchasing behaviors, revealing that the average customer spends **$59.76** per transaction and leaves an average review rating of **3.75**.

Analytical Workflow

1. Data Preparation and Cleaning (Python)

The raw dataset (`customer_shopping_behavior.csv`) was processed using Python's Pandas library to prepare it for analysis.

- **Column Standardization:** Column names were converted to lowercase, and spaces were replaced with underscores for SQL compatibility. The `purchase_amount (usd)` column was simplified to `purchase_amount`.
- **Handling Missing Data:** Missing values in the `review_rating` column were filled using the median rating of the respective product `category`.
- **Feature Engineering:**
    - **Age Groups:** Customers were segmented into four equal quartiles (`age_gp`): *young adult, adult, middle aged, and senior*.
    - **Purchase Frequency:** Categorical purchase frequencies (e.g., 'Weekly', 'Annually') were mapped to a numeric `purchase_frequency_days` column (e.g., 7, 365).
- **Redundancy Removal:** The `promo_code_used` column was dropped because it was perfectly correlated with `discount_applied`.

Code Snippet: Data Cleaning

```
In [28]: df["review_rating"]=df.groupby("category")["review_rating"].transform(lambda x: x.fillna(x.median()))

In [29]: df.isnull().sum()

Out[29]: customer_id              0
         age                      0
         gender                   0
         item_purchased           0
         category                 0
         purchase_amount          0
         location                 0
         size                     0
         color                    0
         season                   0
         review_rating            0
         subscription_status      0
         shipping_type            0
         discount_applied         0
         promo_code_used          0
         previous_purchases       0
         payment_method           0
         frequency_of_purchases   0
         dtype: int64
```

2. Database Integration

The cleaned DataFrame was pushed to an SQL Server database (`customer_behavior`) into a table named `customer_behavior_table` using SQLAlchemy.

3. Exploratory Data Analysis (SQL)

Several business questions were answered using SQL queries to uncover deep behavioral trends:

- **Revenue Demographics:** Total revenue was compared by gender and by age group.
- **Discount & Spending Habits:** Identified customers who used discounts but still spent more than the average purchase amount. Another query calculated the top 5 products with the highest percentage of discount usage.
- **Product Performance:** Identified the top 5 products with the highest average review ratings and ranked the top 3 most purchased products within each category.
- **Customer Segmentation:** Segmented customers based on previous purchases into 'new' (1 purchase), 'returning' (2-10 purchases), and 'loyal' (>10 purchases).
- **Subscription Value:** Analyzed if subscribed customers spend more on average and checked if repeat buyers (>5 purchases) are more likely to subscribe.

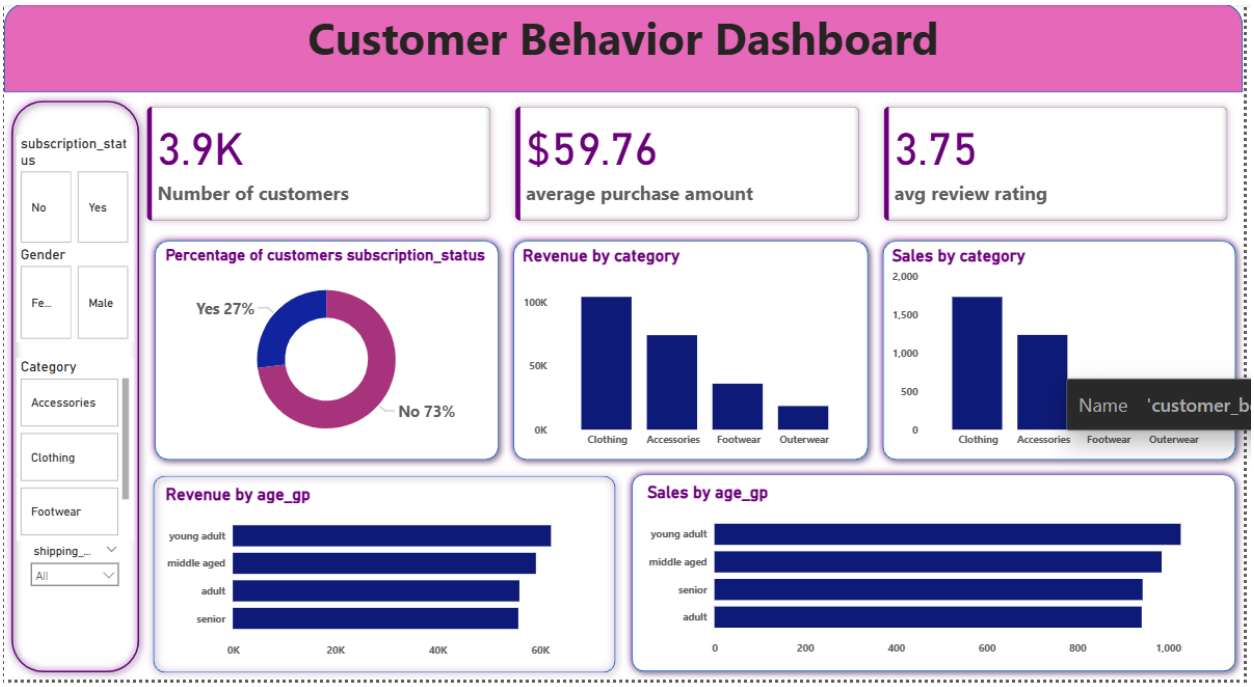
Code Snippet: Customer Segmentation SQL

```sql
--segmentation of customers based on their previos purchases
with customer_type as (
select customer_id,previous_purchases ,
case
when previous_purchases=1 then 'new'
when previous_purchases between 2 and 10 then 'returning'
else 'loyal'
end as customer_segment

from customer_behavior_table

)
select customer_segment,count(*) as 'number of customers' from customer_type
group by customer_segment
```

Dashboard Overview

The **Customer Behavior Dashboard** provides a high-level visual summary of the SQL analysis. It features interactive filters for subscription status, gender, category, and shipping type.

- **Top KPI Cards:** Displays a total of 3.9K customers, an average purchase amount of $59.76, and an average review rating of 3.75.
- **Subscription Breakdown:** A donut chart reveals that **73% of customers do not have a subscription**, while 27% are subscribed.
- **Category Performance (Bar Charts): Clothing** is by far the highest performing category, generating over $100K in revenue and roughly 1,750 sales. This is followed by Accessories, Footwear, and Outerwear.
- **Demographic Performance (Horizontal Bar Charts):** Breaking down revenue and sales by `age_gp` reveals that **young adults** and **middle-aged** customers are the top two demographics for both sales volume and total revenue generation.



Key Insights

1. **Clothing is the Primary Driver:** The "Clothing" category dominates both transaction volume and overall revenue, vastly outperforming Accessories, Footwear, and Outerwear.
2. **Young & Middle-Aged Adults Lead Sales:** The newly engineered `age_gp` feature successfully highlighted that "young adult" and "middle-aged" segments are the most valuable demographic cohorts.
3. **Untapped Subscription Potential:** With **73% of the customer base unsubscribed**, there is massive potential for recurring revenue. The SQL analysis exploring repeat buyers' likelihood to subscribe is highly relevant to closing this gap.

Recommendations

- **Targeted Subscription Campaigns:** Since the vast majority of shoppers (73%) are unsubscribed, launch targeted marketing campaigns aimed at the "loyal" customer segment identified in the SQL analysis (those with >10 previous purchases) to convert them into subscribers.
- **Inventory & Marketing Focus:** Allocate more marketing budget and inventory depth to the **Clothing** category, specifically tailoring the messaging toward **young adult and middle-aged** demographics, as they are proven to convert at higher rates.

- **Leverage Discounts on High-Performing Products:** Utilize the SQL insights regarding products with the highest discount application rates to build strategic bundle offers, encouraging customers to spend above the $59.76 average cart value.